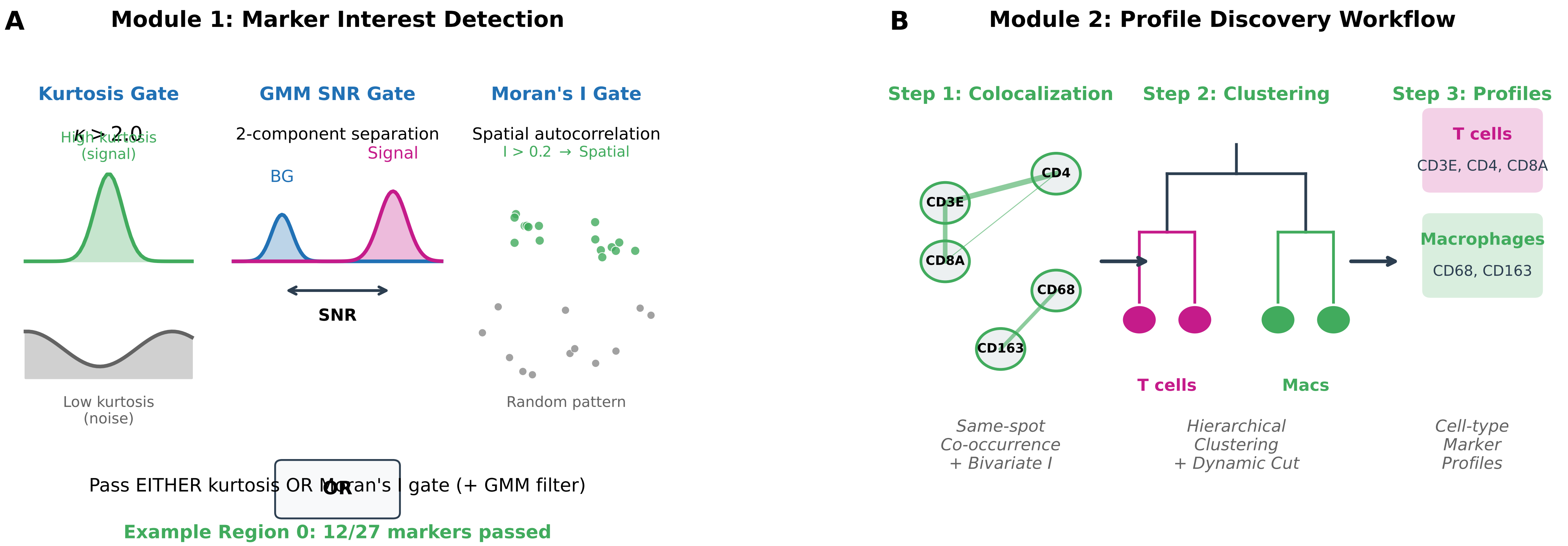

C **Xenium RCC Single-Cell Demonstration**

Dataset: 10x Xenium Human Renal Cell Carcinoma

5 tissue regions | 27 protein markers | ~200K cells

Module 1:

27/27 markers interesting (all regions)

100% sensitivity to critical markers

Module 2a:

351 marker pairs analyzed

100% positive colocalization accuracy

Key Findings:

- Same algorithm at Visium/cell resolution
- CD3E-CD4-CD8A form T cell cluster
- CD68-CD163 form macrophage cluster
- Automatic profile matches ground truth

Resolution-agnostic: single-cell Moran's I replaces spot-level statistics

D **Discovered Profiles vs Known Markers**

Cell Type	Known Markers	Discovered
B cells	CD20	= CD20
CD4+ T cells	CD3E, CD4	= CD3E, CD4
CD8+ T cells	CD3E, CD8A	= CD3E, CD8A
Macrophages	CD68, CD163	= CD68, CD163
Endothelial	CD31	= CD31
Epithelial	PanCK	= PanCK
Fibroblasts	alphaSMA	= alphaSMA

= Exact match + Superset ? Not found